

6.5 Average Value of a Function

Define a function f on the interval $[a, b]$. We wish to create a formula to find the average of n values of f .

Definition

The **average value of f on $[a, b]$** is given by

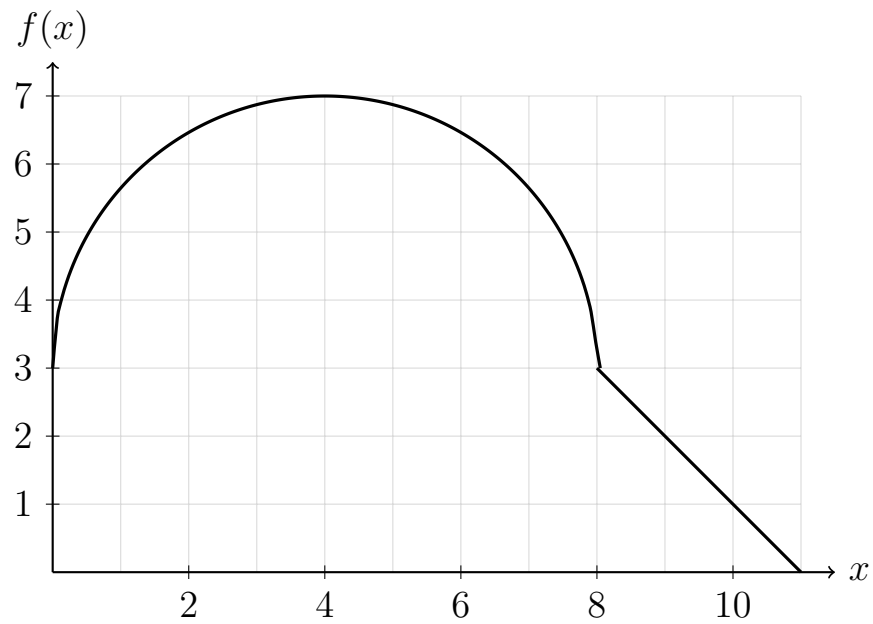
What is a geometric interpretation of the average value of f on $[a, b]$?

Example 1

Find the average value of $f(x) = x \sec(x^2)$ on $[0, \sqrt{\pi/4}]$.

Example 2

Find the average value of f , given in the graph below, on $[0, 11]$.



Example 3

Consider $f(x) = \sin(x) + 2$ on the interval $[0, 2\pi]$.

(a) Find the average value of f on $[0, 2\pi]$.

(b) Plot f on $[0, 2\pi]$. Draw a rectangle with corners, $(0, 0)$, $(2\pi, 0)$, $(0, f_{\text{ave}})$, $(2\pi, f_{\text{ave}})$.

(c) Compare $\int_0^{2\pi} f(x) \, dx$ to the area of the rectangle drawn in part (b).

The Mean Value Theorem for Integrals

Let f be continuous on $[a, b]$, then there exists a number c in $[a, b]$ such that

$$f(c) =$$

Equivalently, $\int_a^b f(x) \, dx =$

Example 4

Let $f(x) = \frac{1}{x}$ be defined on $[1, 3]$.

(a) Find the average value of f on the given interval.

(b) Find c such that $f_{\text{ave}} = f(c)$.